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Large resistive 2D Micromegas with genetic multiplexing and some imaging applications

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ABSTRACT

The performance of the first large resistive Micromegas detectors with 2D readout and genetic multiplexing is presented. These detectors have a 50×50 cm^2 active area and are equipped with 1024 strips both in *X*- and *Y*-directions. The same genetic multiplexing pattern is applied on both coordinates, resulting in the compression of signals on 2×61 readout channels. Four such detectors have been built at CERN, and extensively tested with cosmics. The resistive strip film allows for very high gain operation, compensating for the charge spread on the 2 dimensions as well as the S/N loss due to the huge, 1 nF input capacitance. This film also creates a significantly different signal shape in the *X*- and *Y*-coordinates due to the charge evacuation along the resistive strips. All in all a detection efficiency above 95% is achieved with a 1 cm drift gap. Though not yet optimal, the measured 300 μm spatial resolution allows for very precise imaging in the field of muon tomography, and some applications of these detectors are presented.

1. Introduction and context

Particle and nuclear physics experiments have triggered outstanding progress in particle detectors and instrumentation capabilities over the last 50 years. The combined needs for higher precision, higher flux, smaller material budget have in particular led to the development of robust Micro-Pattern Gaseous Detectors (MPGD) in the mid 1990s [1]. MPGDs can stand very high fluxes thanks to a rapid charge evacuation and a fine granularity of the readout elements, and they are nowadays used in numerous particle physics experiments, see e.g. [2,3] for some recent applications. In practice the granularity level is often determined by the required resolution constraint, and in many cases only a few elements carry a useful signal. From this observation, a multiplexing pattern [4] was developed for strip detectors, to connect a single electronic channel to several strips. The pattern uses as a fundamental assumption that at least two neighbouring strips receive a signal from an incoming particle, thus creating an information redundancy. Combinatorial considerations allow for a systematic connection in which the information lost by the multiplexing is exactly compensated by this redundancy. A first Micromegas [5] prototype built from the bulk [6] technology was tested in cosmics, with 1024 strips connected to only 61 channels. Efficiencies around 90% were reported [4], mainly limited by two effects:

- an unfavourable S/N ratio driven by the huge input capacitance (approximately 1 nF);
- a fraction of events with a cluster size of only one, i.e. failing the fundamental assumption of the multiplexing.

These two limitations can be largely overcome by the additional use of a resistive strip film [7] which allows for significantly higher gains and larger cluster sizes. The corresponding increase of the S/N ratio further suggested that a 2D readout should be possible even if it implies a reduction of at least a factor of two for a given coordinate. A new, 2D layout was therefore designed, and is described in the next section. Section 3 reports on the performance obtained with several prototypes, and Section 4 focuses on imaging applications of this innovative detector.

2. Description of the prototypes and experimental setup

2.1. Prototypes

The 2D version is a natural extension of the 1D prototype, combined in *X* and *Y* thanks to the resistive strip technology.

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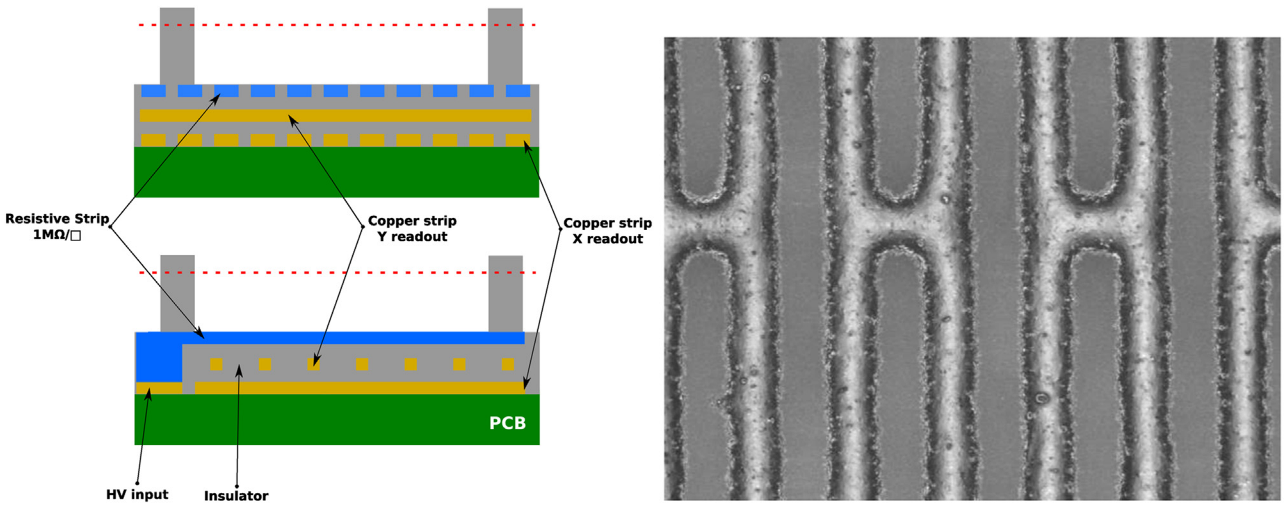

Fig. 1. Left: layout of the resistive detectors used. Right: picture of the resistive strips with their interconnecting ladders.

The resulting layout consists of the following layers (see Fig. 1 (left)):

- the micro-mesh;
- a layer of resistive, X strips, with a 488 μm pitch and 380 μm strips;
- a layer of readout Y strips with the same pitch, and thin, 100 μm strips;
- a layer of readout X strips with the same pitch, and 380 μm strips.

Thinner Y strips allow for a significant signal to be induced on the most inner X strip plane. The multiplexing pattern is the same on both coordinates, and is identical to the 1D prototype. The first resistive detectors were originally prepared with a resistive ink spread on the PCB, and manually polished to achieve the final desired resistivity. Though this lengthy method was successfully used both at CERN and in industry (ELVIA), it is now replaced by a serigraphic process in which the resistive ink is applied on a Kapton foil through a stencil. This process has the advantages to be more reproducible, with larger production capabilities and to be independent on the bulk manufacturing. The Kapton foil is indeed prepared separately, and pressed on the PCB prior to the first lamination with a 50 μm thick glue. In our case the resistive films were manufactured at CERN, and interconnecting *ladders* were added between the strips as seen in Fig. 1 (right) to ensure a better homogeneity of the resistivity.

Finally, a relatively large, 1 cm drift gap was chosen to reach a comfortable efficiency plateau and also to implement micro-TPC algorithms in future studies. Four such detectors (MG2D) with an active area of $50 \times 50\ \text{cm}^2$ have been initially manufactured at CERN. The printed circuit board of another detector was manufactured by the ELVIA group who also pressed the resistive film on it, and the embedment of the micromesh on top of the PCB was finalized at CERN.

2.2. Experimental setup

All the prototypes were placed in the cosmic test bench developed for the CLAS12 project [8,9]. The coincident signals from two $60 \times 60\ \text{cm}^2$ plastic scintillators located at the extremities of the bench provided the trigger for the readout electronics. Additional CosMulti Micromegas [4] were installed between the scintillators to reconstruct the muon tracks, but these detectors were progressively replaced by the MG2D prototypes once characterized and validated. All these reference detectors have a $50 \times 50\ \text{cm}^2$ active area with 1024 strips, corresponding to a 488 μm pitch. At least three 2D reference detectors should detect a signal with a residual below 2 mm to reconstruct the muon track. The comparison between the extrapolated muon positions and the recorded positions of the test detector then provides its 2D efficiency and its spatial resolution. For these measurements we switched from the T2K/AFTER [10] to the CLAS12/DREAM [11] electronics, which is now available and was designed specifically for relatively large input capacitances. Studies with a Fe^{55} source placed on a detector alternatively equipped with AFTER and DREAM electronics showed that the S/N ratio is increased by more than 20% in the latter case. The signal transmission between the detectors and the electronics was ensured by 2 m Hitachi micro-coaxial cables, adding a negligible 100 pF capacitance. In the final configuration with the four prototypes, a single 8-connector Front End Unit with only 512 electronics channels was needed to read all the signals of four MG2D. Detectors were flushed with Argon-Isobutane mixture (95–5%) with a total flux of 5 L/h parallelized in two doublets.

3. Results and performance

The readout electronics provides the signal amplitude on each channel every 48 ns. For the current analysis, we recorded 32 samples, corresponding to a time window of around 1.5 μs. Only signals with an amplitude higher than five times the noise level (measured with a dedicated run using a random trigger) are considered for the clustering. Clusters of consecutive strips are then formed, and their positions in each detector are calculated from the barycentre of the recorded signals. As explained in the previous section, muon tracks are then reconstructed in the reference detectors and compared with the signals of the test MG2D. Only the events for which the residual is less than 2 mm are kept for analysis. With an average (reconstructed) muon rate of around 3 Hz, a one hour data acquisition already provides enough of statistics for the determination of spatial resolution and efficiency.

From the raw data, the first visible feature of the MG2D detectors is an asymmetry between X and Y signals due to the orientation of resistive strips. Y clusters indeed are much larger because of the slow, perpendicular charge evacuation. This effect is particularly visible in Fig. 2, showing the time development of the signal. The maximum amplitude of the central strip of the Y cluster (in blue) occurs around sample 12. The charges then evacuate on the sides, and induce a delayed signal on the neighbouring strips,

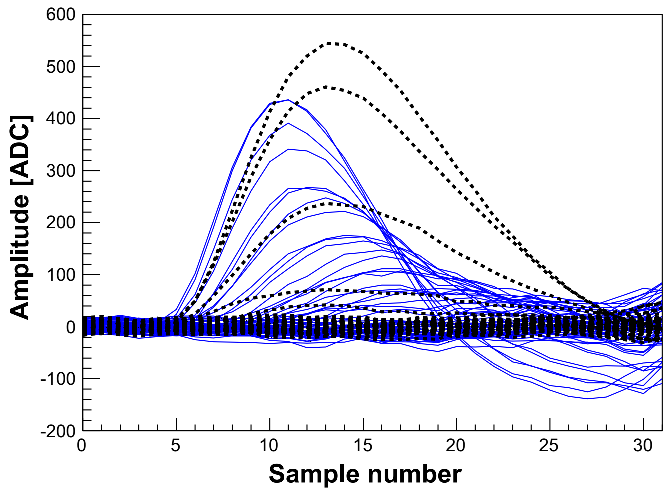

Fig. 2. Event display on Y (blue lines) and X (black dotted lines) coordinates of an MG2D detector in the cosmic test bench. Pedestals and common mode have been subtracted. The sampling frequency is 20.8 MHz. (For interpretation of the references to colour in this figure caption, the reader is referred to the web version of this paper.)

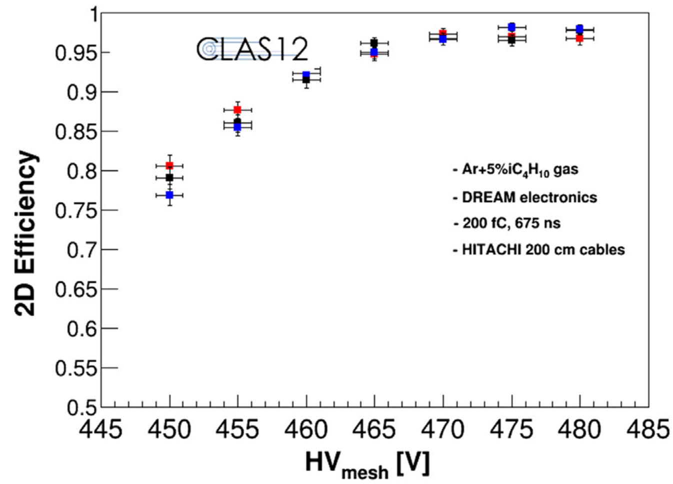

Fig. 3. 2D efficiency curves for 3 MG2D detectors, measured in the cosmic test bench. Errors bars are statistical.

while an undershoot develops on central strips as the electrons move away.

The 2D efficiency curves are displayed in Fig. 3 for three detectors. This 2D efficiency is estimated with a reference muon sample determined as above, and counting the muon fraction seen simultaneously by the X- and Y-coordinates. For these measurements, the drift high voltage is fixed at 800 V, ensuring a practically 100% micro-mesh transparency. In spite of the charge sharing, a comfortable plateau is reached for all the detectors thanks to the resistive technology and the new electronics. When the micromesh high voltage reaches around 470 V, the signal amplification is large enough to yield a full efficiency. The inter-strip distances and layer thickness, inspired from Atlas-NSW[1] prototypes [12] allows both coordinates to receive approximately the same charge, and thus to have the same efficiency. This was not the case for the first ELVIA prototype where thicker Copper and glue layers led to a smaller signal on X readout strips.

The 2D efficiency combining X- and Y-detection is significantly higher than the product of 1D efficiencies (corresponding to the muon fraction detected by only one coordinate) thanks to a strong correlation between the two signal, illustrated in Fig. 4 (left). Stable 2D efficiencies above 95% were routinely measured during several months of operation. Last but not least, spatial resolution obtained from the residual (difference between extrapolated track position and measured position in the test detector) distribution was found to be relatively large, of the order of 300 μm (see Fig. 4 (right)). In this figure, a two-Gaussian fit has been used, to take into account a tail presumably coming from dead strips in the detectors splitting the clusters and degrading the spatial resolution. Though misalignments clearly remain in the mechanical structure of the cosmic bench, several detectors elements can be improved. In particular, a second version has been built recently without interconnecting *ladders* and with an improved mechanical frame to ensure better planarity.

4. Imaging capabilities with muons

As already reported [4], genetic multiplexing can be considered for relatively high flux experiments, thanks to the flexibility in the degree of multiplexing. However it clearly offers more applications at low flux thanks to the stronger channel reduction, and in particular in the field of muon tomography. This technique consists in imaging an object using naturally available cosmic rays. Depending on the size of the object and its configuration, imaging can be realized either on absorption or deviation mode.

The former case relies on the muon flux measurement in a given direction to determine the matter opacity along this line. It is known for decades, see e.g. [13,14], and has found in the last 15 years several applications, especially in volcanology [15]. So far the dominant technology used is based on scintillator telescopes which are very robust but offer limited spatial resolution, of the order of 1 cm. In spite of long telescopes the angular resolution is poor and limits the imaging quality. On the contrary emulsion-based telescopes provide resolutions at the micron level, but cannot provide dynamic imaging [15] while being very sensitive to the environment (temperature, humidity). Furthermore, such a small resolution is often useless in practice because of the natural blurring induced by the interactions of muons within the object or in the air. A robust, compact telescope offering submillimetric resolution with dynamic imaging capabilities thus represents a very interesting compromise in this field, and is still lacking.

The deviation mode is based on the measurement of muon deflection in matter due to the Coulomb scattering. In this case the muon trajectory is reconstructed upstream and downstream the object. Assuming a single (significant) interaction of the muons with this object, the intersection of the two half-trajectories gives access to the position of this interaction, while the typical angle (estimated with many muons) provides information on the local density. The typical deviation angle of a 4 GeV muon in a 1-cm Lead (resp. Iron) brick yields roughly 4.7 mrad (resp. 2.6 mrad). A competitive imaging setup should therefore reach angular resolution well below these values to be performant.

To check their performance for muon imaging, our four detectors were assembled in a small, transportable Aluminum-based structure of 1.5 m height from the Norcan company. Two doublets were formed, separated by an Aluminum plate where Lead bricks can be arranged. The distance between the two detectors of a doublet can be varied from 5 to 40 cm, giving access to sub-mrad deviation measurements with a compact instrument. Fig. 5 illustrates the imaging capabilities of this setup as a function of time for a Lead square. One can see in particular that a 20-min run is enough to reconstruct the Lead pattern. To our knowledge, there is no other setup which offers such precise and fast imaging capabilities with muons while being compact enough to be transported easily. This

[1] New Small Wheel.

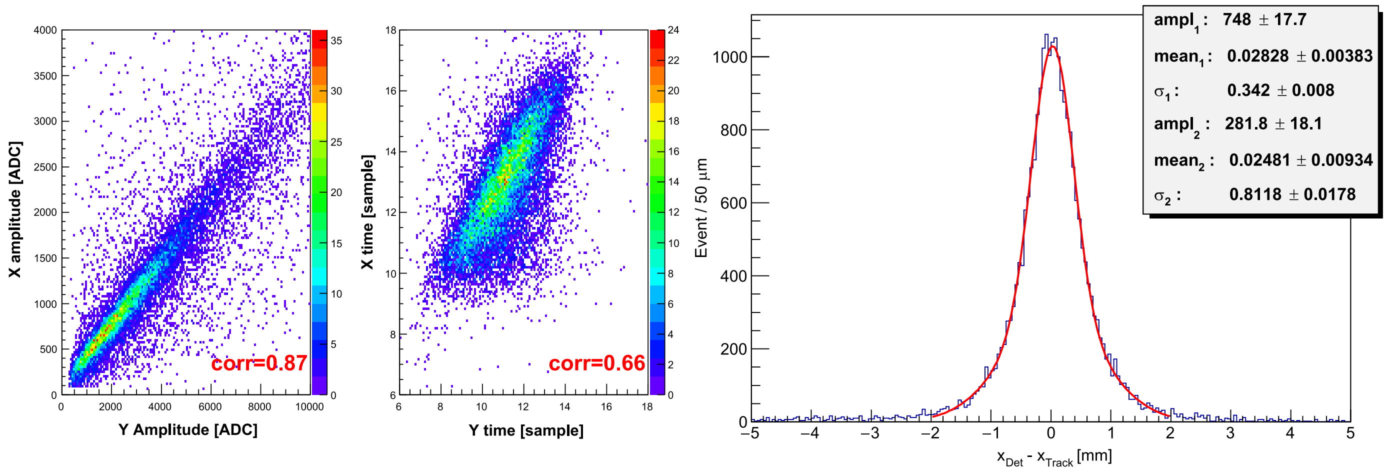

Fig. 4. Correlation in the cluster amplitude (left) and in the time of the signal maximum (middle) for *X*- and *Y*-coordinates of an MG2D. Right: typical residual distribution obtained in the detectors, corresponding to a spatial resolution around 300 μm.

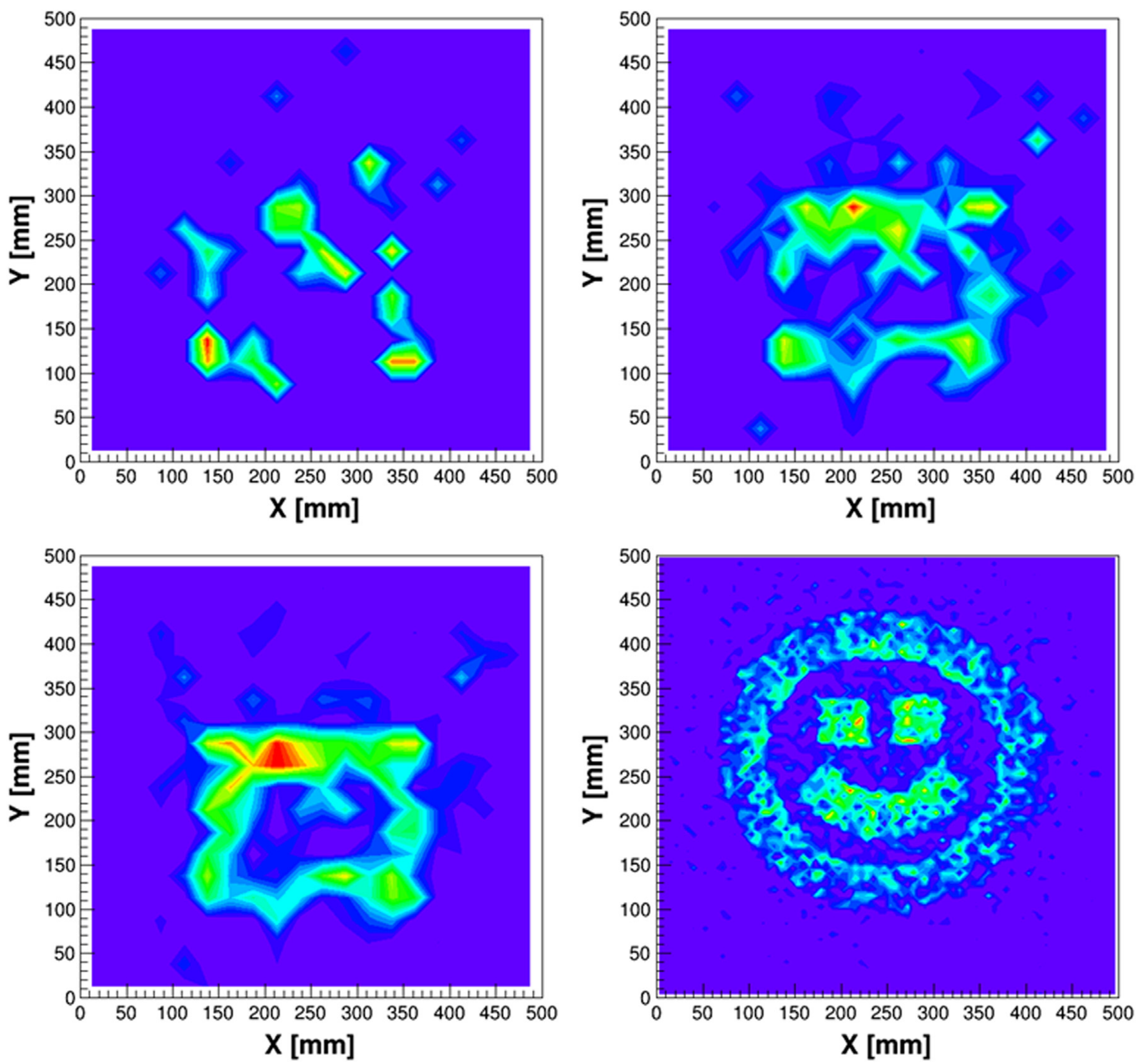

Fig. 5. Reconstruction of Lead brick patterns with the TomoMu setup: a simple square seen after 2 (top left), 10 (top right) and 20 (bottom left) minutes, and a smiley (bottom right) after 1 h of data. The plots show the distributions of the intersection points of the two half-trajectories (POCA algorithm), where each point is weighted by the corresponding deviation angle.

setup, called TomoMu and partly funded by the *Diagonale Paris-Saclay*, will soon be available for communication purpose and exhibitions. The audience will first be asked to place the bricks on the plate, and will see the image appearing online while a speaker will explain the principles of the instrument—cosmic rays, particle detection, interaction of particles with matter. It is also foreseen to use

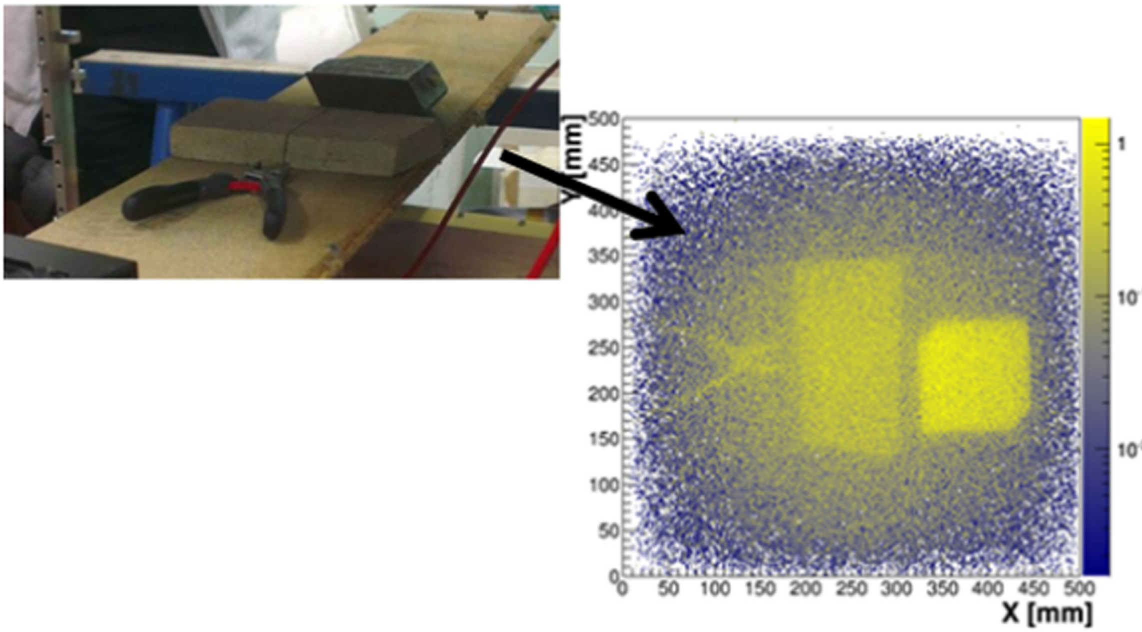

Fig. 6. Reconstruction of different objects in the TomoMu setup, after 2 days of data: cutting pliers, a limestone and a Lead brick. The metallic structure inside the limestone was also seen. On the picture, even the wooden board on which objects were disposed is visible.

such an instrument for M2 students during laboratory works to illustrate the concepts studied during instrumentation lectures.

The imaging capabilities of this instrument was further checked with lower Z material, as illustrated in Fig. 6. In this case the acquisition time is longer, of the order of a few days, but allows for the imaging of much less dense materials. This instrument has raised the interest of the French LRMH,[2] for example to image in situ statues or walls and reveal the position of internal metallic structures. In addition to the natural origin of cosmics, the excellent spatial resolution of these detectors combined with the reduced electronics indeed allows for particularly compact, transportable, and low consumption scanners.

The four detectors were recently assembled in a muon telescope to evaluate their performance in the absorption mode. This telescope has been operated at Saclay to provide static and dynamic images of a Water Tower, and results will be presented soon in a separate paper.

5. Conclusion

First large area resistive 2D multiplexed Micromegas have been built and successfully tested with cosmics. These detectors dedicated to low flux experiments show very good efficiency and stable performance in time. Though further studies are needed to improve the spatial/angular resolution of the apparatus, the 2D localization of particles with a 300 μm precision on such large surface with only 122 channels is a remarkable feature of this detector, and several applications in and beyond particle physics are already initiated. Their use in the challenging field of muon tomography have been investigated with very promising results. Last but not least, large scale, industrial applications are now at hand thanks to the know-how transfer successfully completed with the ELVIA company.

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